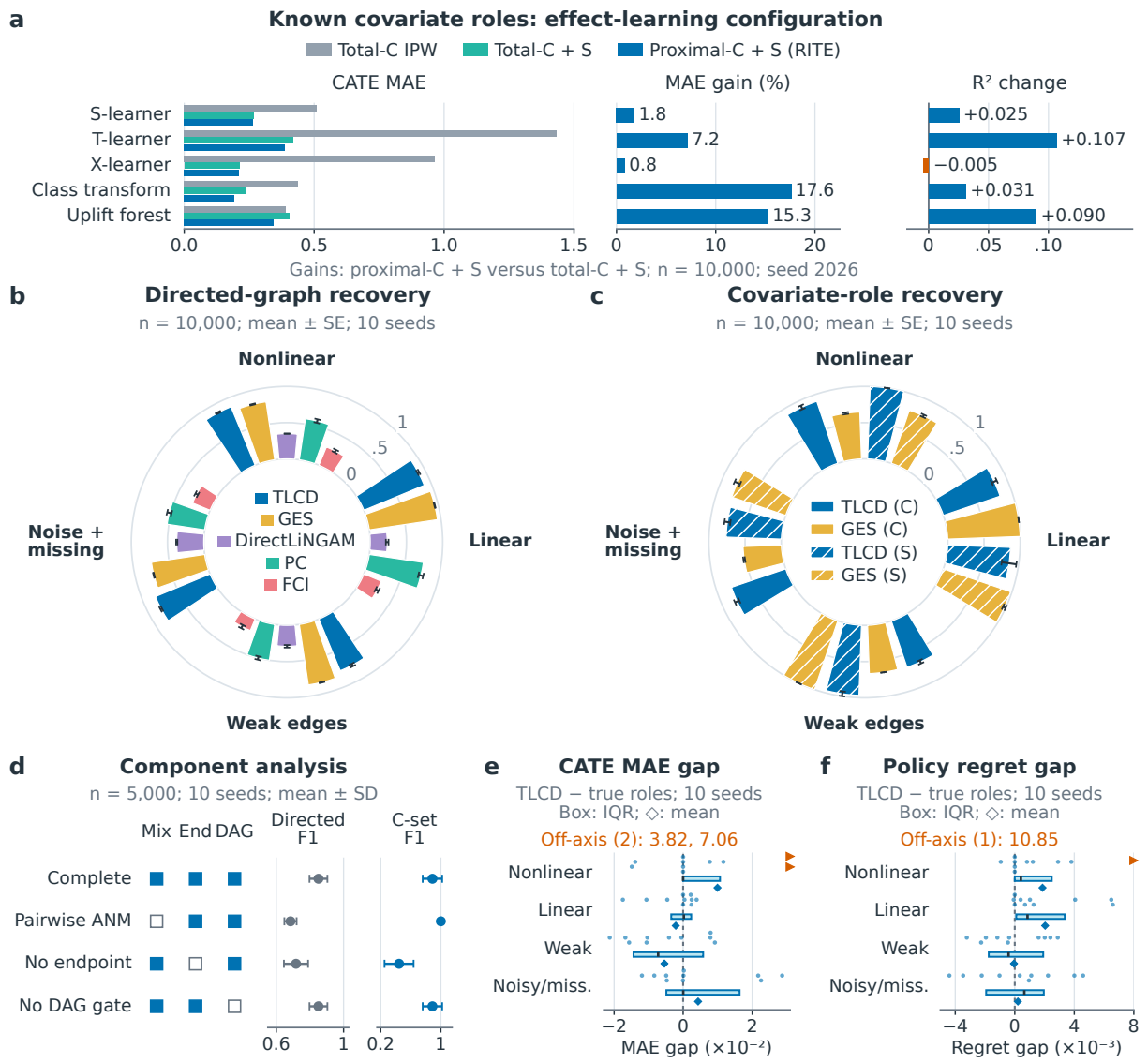

a
Known covariate roles: effect-learning configuration
Total-C IPW
Total-C + S
Proximal-C + S (RITE)
CATE MAE
MAE gain (%)
R² change
S-learner
T-learner
X-learner
Class transform
Uplift forest
0.0
0.5
1.0
1.5
1.8
7.2
0.8
17.6
15.3
0
10
20
+0.025
+0.107
−0.005
+0.031
+0.090
0
.05
.10
Gains: proximal-C + S versus total-C + S; n = 10,000; seed 2026
b
Directed-graph recovery
n = 10,000; mean ± SE; 10 seeds
Nonlinear
Noise + missing
Linear
Weak edges
1
.5
0
TLCD
GES
DirectLiNGAM
PC
FCI
c
Covariate-role recovery
n = 10,000; mean ± SE; 10 seeds
Nonlinear
Noise + missing
Linear
Weak edges
1
.5
0
TLCD (C)
GES (C)
TLCD (S)
GES (S)
d
Component analysis
n = 5,000; 10 seeds; mean ± SD
Mix
End
DAG
Directed F1
C-set F1
Complete
Pairwise ANM
No endpoint
No DAG gate
0.6
1
0.2
1
e
CATE MAE gap
TLCD − true roles; 10 seeds
Box: IQR; ◇: mean
Off-axis (2): 3.82, 7.06
Nonlinear
Linear
Weak
Noisy/miss.
−2
0
2
MAE gap (×10⁻²)
f
Policy regret gap
TLCD − true roles; 10 seeds
Box: IQR; ◇: mean
Off-axis (1): 10.85
Nonlinear
Linear
Weak
Noisy/miss.
−4
0
4
8
Regret gap (×10⁻³)